Birla Institute of Technology and Science, Pilani.

Mid-Semester Examination: EEE F243/ INSTR F243: Signals and Systems

Marks: 50 **AY: 2015-16, Semester: II** **Date: 15-March-2016, Tuesday**

Time: 90 minutes **Maximum Marks: 50** **Pages: 02**

Note: Neat and legible figures must be drawn wherever mentioned with all credentials.

1 **10**

$$x(t) = \begin{cases} t+5, & -5 \leq t \leq -4 \\ 1 & -4 \leq t \leq 4 \\ 5-t & 4 \leq t \leq 5 \\ 0, & otherwise \end{cases}$$

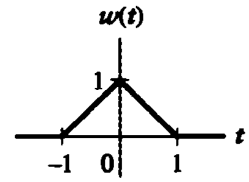

Figure (1a).

a. Sketch the signal $x(t)$
b. Sketch $y(t) = x(at)$; for $a = 5$ and $a = 0.2$
c. Sketch $v(t) = x(10t - 5)$
d. Determine the total energy of $x(t)$
e. Sketch $m(t) = \sum_{k=-5}^{5} w(t - 2k)$; $w(t)$ is shown in figure (1a).
f. Is the signal $m(t)$ periodic or aperiodic? If periodic find the fundamental period. If aperiodic, what condition will make the signal periodic and what will be the fundamental period.

2 a. Consider an LTI system whose response to the signal $x_1(t)$ in figure (2a) is the signal $y_1(t)$ illustrated in figure (2b). Determine and sketch carefully the response of the system to the input $x_2(t)$ depicted in figure (2c) and input $x_3(t)$ shown in figure (2d). **12**

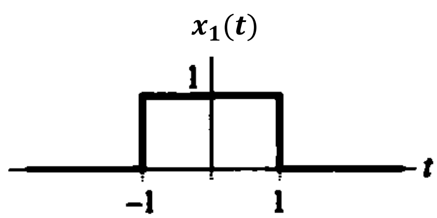

Figure (2a)

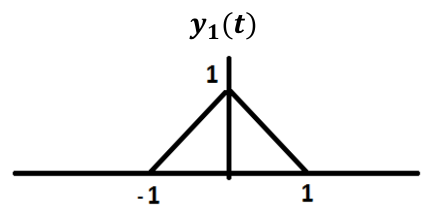

Figure (2b)

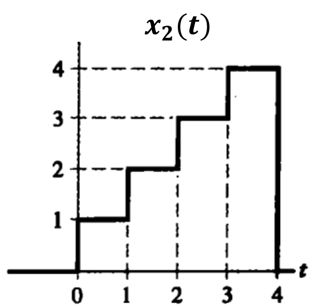

Figure (2c)

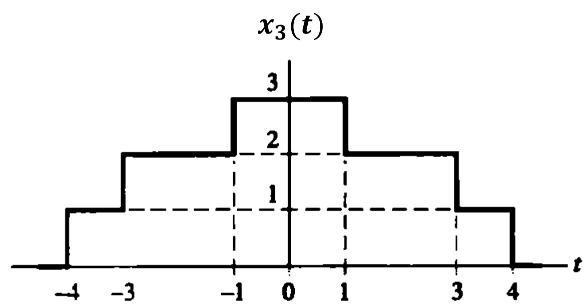

Figure (2d)

b. Determine and sketch $m(t) = x(t) * g(t)$ and $p(t) = x(t) * f(t)$, where

$$f(t) = \begin{cases} 1 & -\pi \leq t \leq \pi \\ 0 & otherwise \end{cases}$$

$$g(t) = u(t)$$

$$x(t) = \begin{cases} \sin(t) & 0 \leq t \leq 2\pi \\ 0 & otherwise \end{cases}$$

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3 Consider a system $y(t) = |x(t)|$. **12**

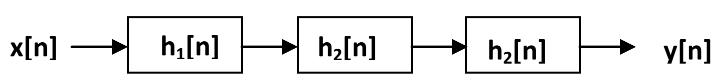
a. Sketch the input and output waveforms if $x(t) = \cos(t)$. What are the fundamental period of the input and output?
b. If $x(t) = \cos(t)$, determine and sketch the Fourier series coefficients of the output $y(t)$ $from$ $(-3$ to $+3)$.
c. What is the amplitude of the dc component of input and output signals.
d. Let $x[n] = sin\left(\frac{3\pi n}{4}\right)$ and $y[n] = \begin{cases} 1, & 0 \leq n \leq 3 \\ 0, & 4 \leq n \leq 7 \end{cases}$ be two signals that are periodic with period 8. Find the Fourier series representation for the periodic convolution of these signals.

4 A person receives an automobile loan of Rs. 1,00,000/- from bank at the interest rate of 1% per month. The loan is to be retired by equal monthly payment of Rs. D. Use first order difference equation to determine the monthly payment (Rs. D) required to pay-off the loan in 30 Years (360 payments) and 15 years (180 payments) **06**

5 Consider the cascade interconnection of three casual LTI systems, illustrated in figure (5a). The impulse response $h_2[n]$ is $h_2[n] = u[n] - u[n-2]$, and overall impulse response is $h[n] = \{\underset{\uparrow}{1}, 5, 10, 11, 8, 4, 1\}$. $\uparrow$ indicates the value at n=0. **10**

a. Find and sketch the impulse response $h_1[n]$
b. Find and sketch the response of overall system to the input $x[n] = \delta[n] - \delta[n-1]$

Figure (5a)



A discrete-time system is both linear and time-invariant. Suppose the output due to an input $x[n] = \delta[n]$ is given in figure (5b).

a. Find and sketch the output due to an input $x[n] = \delta[n-1]$
b. Find and sketch the output due to an input $x[n] = 2\delta[n] - \delta[n-2]$
c. Find and sketch the output due to an input shown in figure (5c)

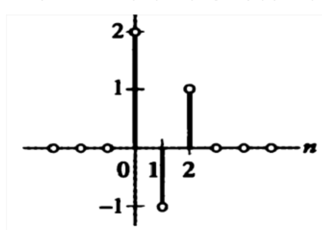


Figure (5b)

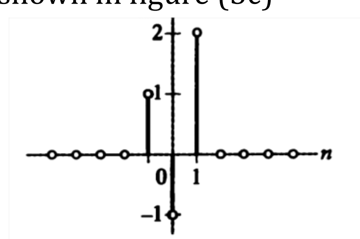


Figure (5c)